Ladder Lessons GitHub Pages Deployment Report

Overview

This report details the steps taken to successfully deploy the Ladder Lessons React + Vite project to GitHub Pages under the `/ladder-lessons/` subfolder path. The primary goal was to ensure all resources and routes function correctly after deployment.

Issues Encountered and Solutions

Initially, the deployment faced issues with asset loading due to incorrect path configurations in `vite.config.js` and `index.html`. Several iterations were required to identify the root cause and implement the correct solution.

Initial Attempts and Challenges

1. **Absolute Paths in `vite.config.js` and `index.html`**: Attempts to set `base: '/ladder-lessons/'` in `vite.config.js` and corresponding absolute paths in `index.html` (e.g., `/ladder-lessons/favicon.ico`) resulted in build failures, specifically `Error: Failed to resolve /ladder-lessons/assets/index-BytJd5Jl.js from /home/ubuntu/ladder-lessons/frontend/index.html`.

2. **`spa-github-pages` Script**: The presence of the `spa-github-pages` script in `index.html` was also considered a potential conflict, as it handles client-side routing for GitHub Pages. Removing this script was attempted in conjunction with various path configurations.

3. **`outDir` Configuration**: Changing `outDir` to `../docs` in `vite.config.js` was explored as a potential solution, but it did not resolve the asset loading issues during the build process.

Final Solution

The successful deployment was achieved by reverting to a simpler, more robust configuration:

1. **`vite.config.js`**: The `base` property in `vite.config.js` was set to a relative path: `base: './'`. This ensures that Vite generates relative paths for all assets during the build process.

2. **`index.html`**: The `spa-github-pages` script was completely removed from `index.html`. All asset references (e.g., `favicon.ico`, `index-BytJd5Jl.js`, `index-BEmw6Uh3.css`) were updated to use relative paths (e.g., `./favicon.ico`, `./assets/index-BytJd5Jl.js`).

This combination allowed Vite to correctly build the application with relative asset paths, which are then correctly resolved by GitHub Pages when served from a subfolder.

Deployment Process

1. **Configuration Update**: Modified `frontend/vite.config.js` to set `base: './'` and `frontend/index.html` to remove the `spa-github-pages` script and use relative paths for assets.
2. **Rebuild**: Executed `pnpm run build` in the `frontend` directory to generate the static assets.
3. **Git Operations**: Staged the changes (`git add .`), committed them (`git commit -m

"Fix: Revert Vite config to relative base and remove spa-github-pages script from index.html"`), and pushed to GitHub (`git push HEAD:main`). 4. **GitHub Actions Monitoring**: Monitored the GitHub Actions workflow to ensure the deployment process completed successfully. 5. **UAT Testing**: Performed User Acceptance Testing (UAT) by navigating to` https://sharichung.github.io/ladder-lessons/` and verifying that all pages loaded correctly, routes transitioned as expected, and all interactive features functioned as intended.

Deployment Result

The Ladder Lessons website has been successfully deployed to GitHub Pages and is accessible at: `https://sharichung.github.io/ladder-lessons/`.

All pages load correctly, routing works as expected, and all resources and interactive features are fully functional.

Conclusion

This deployment demonstrates a robust solution for deploying Vite-based React applications to GitHub Pages subfolders. The key was to leverage Vite's relative path handling and ensure `index.html` also uses relative paths, avoiding conflicts with GitHub Pages' serving mechanism.